

Home Credit Default Risk: Business Problem Statement

Prepared for Home Credit executive leadership | Capstone project | 16 September 2026

Business Situation and Problem

After several divestments and ownership changes since 2022, Home Credit now reports operations in four countries. With a smaller geographic footprint, underwriting quality in the remaining markets is an increasingly important lever for risk-adjusted growth and responsible access to credit for applicants with little or no conventional credit history. Two errors destroy value: declining applicants who would repay limits revenue and financial inclusion, while approving applicants who cannot repay increases credit losses, collection costs, and customer harm. The 2018 Kaggle data provide application, credit-bureau, and prior-loan history for testing risk-ranking methods, but they do not reveal Home Credit's current scorecard, policy thresholds, unit economics, geography, currency, or observation period. The business need is to determine whether an independently validated risk ranking can support safer, more inclusive credit policy before any operational change is considered.

Business Objective and Expected Benefit

The project will estimate each applicant's probability of payment difficulty and translate those estimates into policy scenarios executives can compare. The intended benefit is to serve more applicants likely to repay at an agreed risk level, or reduce payment difficulties at the same simulated selection rate, while preserving responsible lending and improving risk-adjusted value per application.

Proposed Success Measures

The development sample contains 307,511 issued-loan records; 8.1% involved payment difficulty. This is a historical sample outcome rate—not Home Credit's current portfolio default rate or approval rate.

- **Credit quality:** on held-out records, achieve at least a 5% relative reduction in the observed payment-difficulty rate at the same simulated selection rate as a transparent benchmark.
- **Responsible access:** at the same held-out payment-difficulty rate, increase simulated selection of borrowers with no observed payment difficulty by at least 3% relative to the benchmark.
- **Financial value:** after Finance supplies revenue, funding, loss-given-default, and collection-cost assumptions, show positive incremental expected contribution per 1,000 evaluated applications in the base case and no loss under the agreed adverse sensitivity.
- **Analytical readiness:** achieve an out-of-sample risk-ranking score (ROC AUC) of at least 0.78, verify that calibrated risk bands track observed outcomes, and report group selection and error-rate gaps. A gap above 5 percentage points is a proposed internal trigger for Compliance review, not a legal safe harbor.

Analytics Approach and Deliverables

The team will compare an interpretable logistic benchmark with more flexible classifiers using only information available when an application is evaluated. Validation will focus on unseen records, probability calibration, stability, and the business trade-off between approvals and payment difficulties. The final deliverable will include a validated risk-ranking prototype, plain-language driver explanations, threshold and sensitivity scenarios, and an executive recommendation to pilot, revise, or stop. Three risk bands will carry recommended actions for the Credit Risk policy team, not automatic approval or denial, and Credit Risk will retain decision authority.

Scope and Constraints

In scope are offline analysis of the Kaggle application, bureau, and previous-loan tables; data quality treatment; model comparison; validation; and approval-risk scenarios. Out of scope are production integration, new data collection, pricing or loan-limit changes, collections, and individual lending decisions. Released for the 2018 competition, the data are anonymized and historical, with missing and sentinel values, an imbalanced outcome, and no confirmed geography, currency, current policy baseline, or profit-and-loss assumptions. Outcomes are available only for issued loans, so the data cannot directly measure current rejected applicants or the causal effect of a new policy. Demographic and socioeconomic variables may also encode bias. Current portfolio data and review by Credit Risk, Finance, Privacy, and Compliance are required before a pilot.

Execution and Approval

The student analytics team will deliver the business/data audit by 22 September 2026, benchmark and validation by 29 September, value and governance scenarios by 6 October, and the executive recommendation by 13 October. The executive sponsor and leaders from Credit Risk, Finance, Operations, and Compliance will approve a limited pilot, request revision, or stop the initiative.